Amiga KickSmash 32 - In-system smart flash EEPROM

Rev 10.1 2026-10-06

U1
STM32F107VCT6

STM32F107VCT6
72 MHz 32-bit ARM Cortex-M3
64 KB SRAM, 256KB Flash,
USB, ADC, USART, Timer
107 = Connectivity
V = 100 pins
C = 256 KB flash
T = LQFP
6 = –40 to 85 °C
7 = –40 to 105 °C

U9
GD32F105VCT6

Program and debug

H4
Console
GND
CONS_TX
CONS_RX

H6
STM32 SWD
V3P3
STM_SWCLK_TCK
GND
STM_SWDIO_TMS
STM_NRST
STM_SWO_TDO

DFU

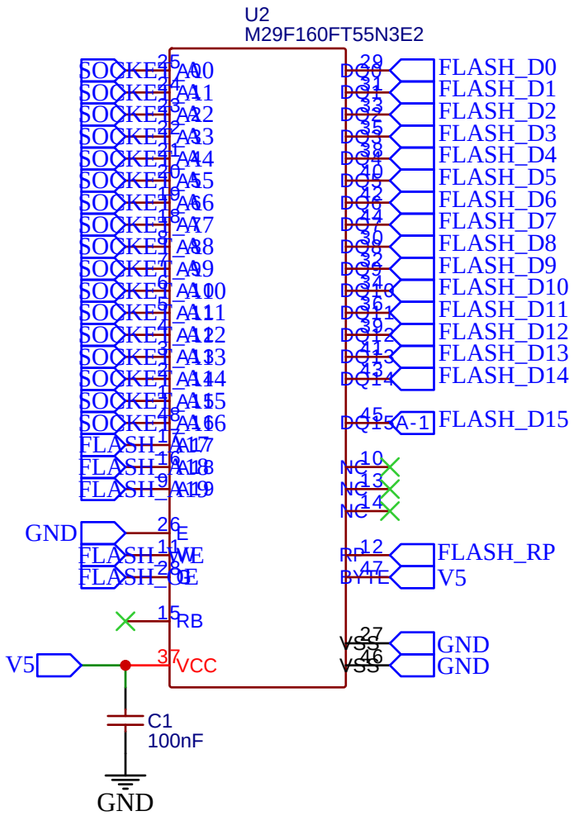
SW1
TS-1088-AR02016
V3P3
STM_BOOT0
R6
5.1K
GND
BOOT0

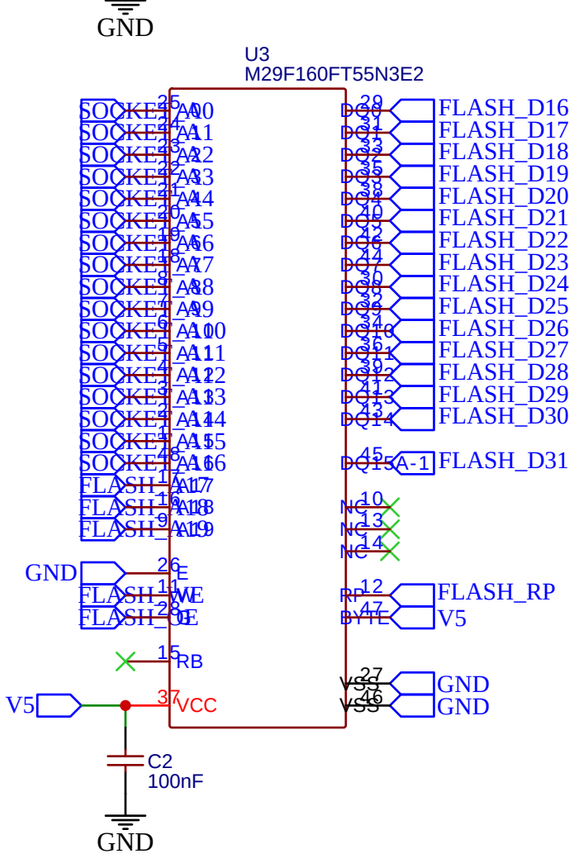
CPU 8M Clock

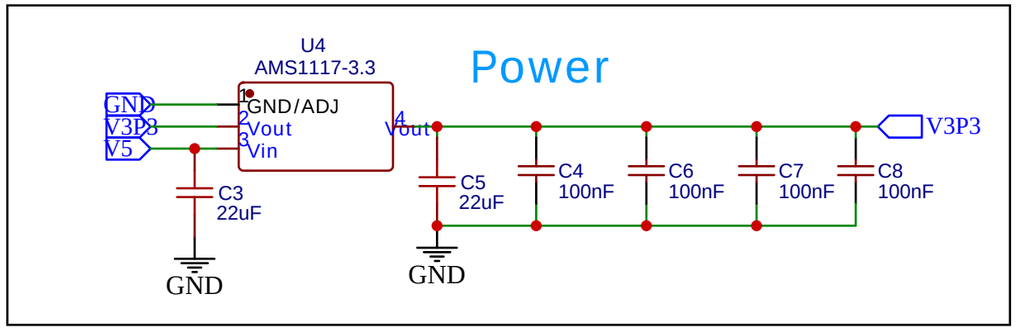
OSC_IN
X1
8MHz
OSC_OUT

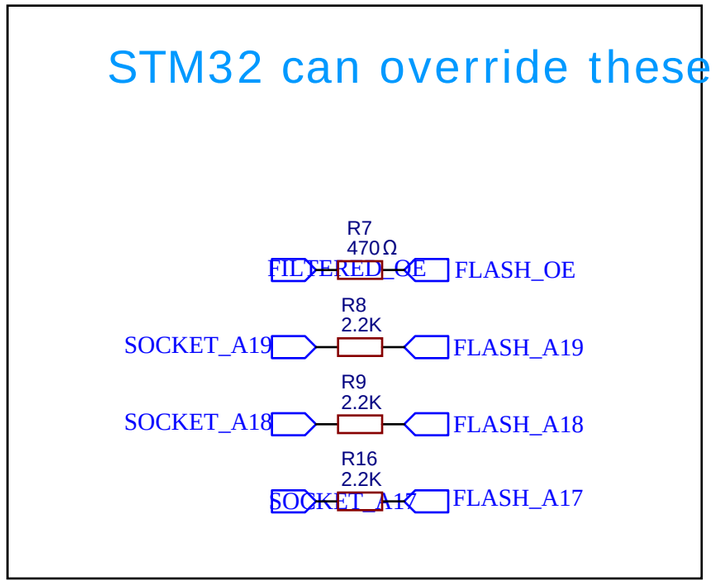
AMIGA_KBRST
KBRST
AMIGA_KBRST
KBRST

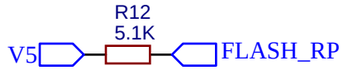
TITLE: STM32
REV: 10.1
Company: eebugs.com
Sheet: 1/4
Date: 2026-10-06
Drawn By: cdhooper
EasyEDA

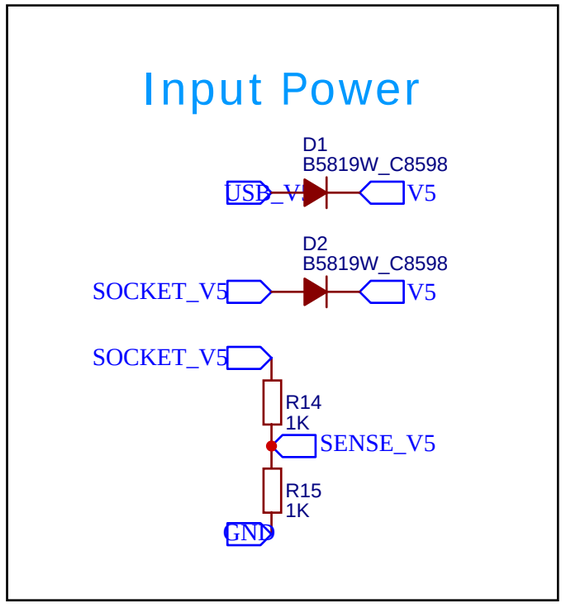


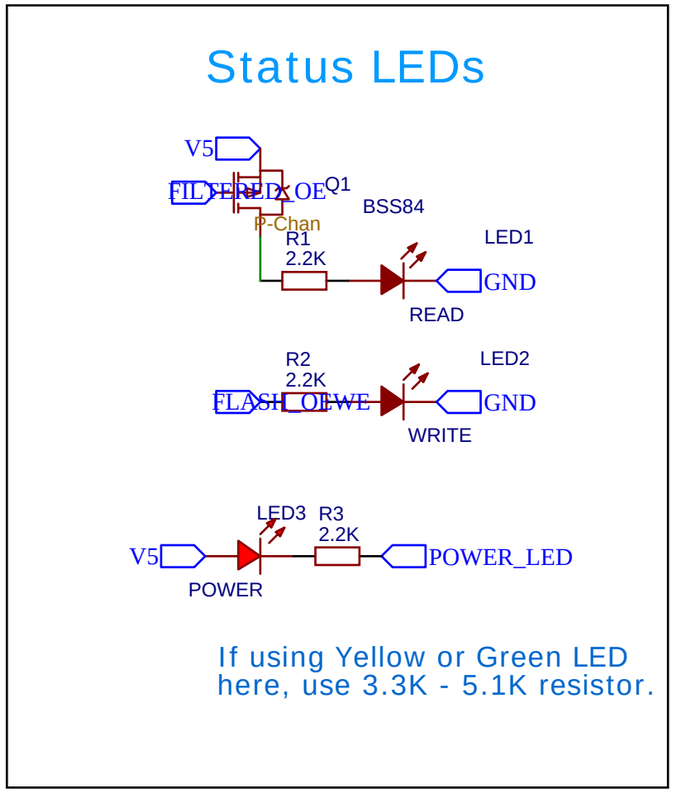






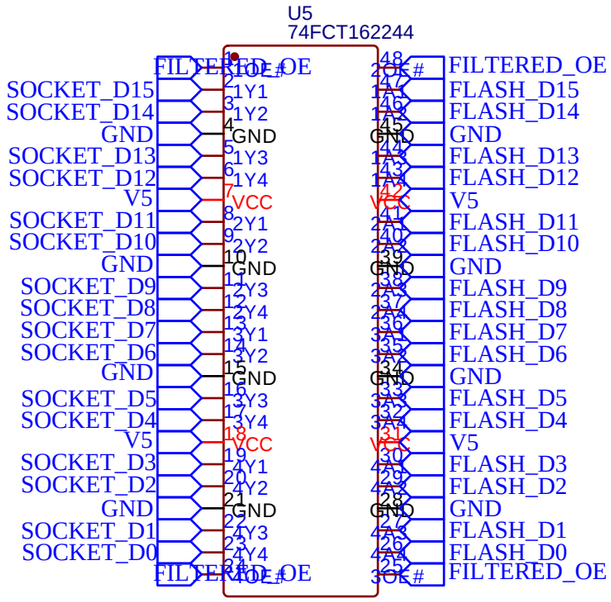


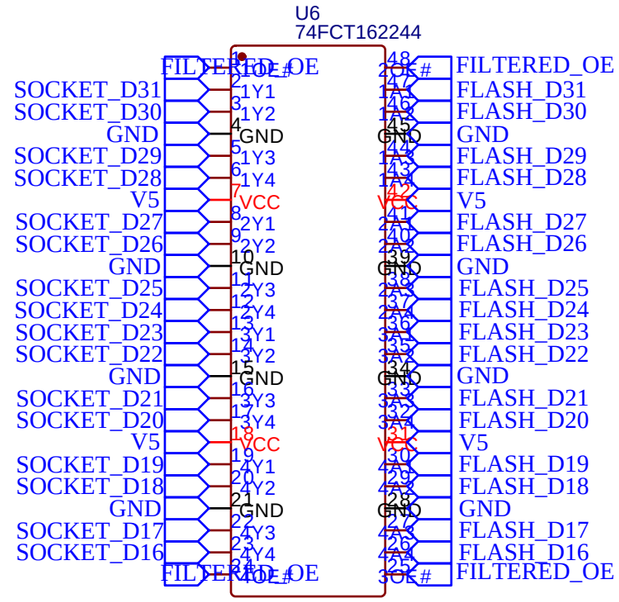




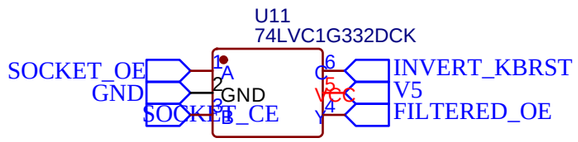
TITLE: Flash + Power		REV: 10.1
EasyEDA	Company: eebugs.com	Sheet: 2/4
	Date: 2026-10-06 Drawn By: cdhooper	

Bus drivers are used to decouple STM32 putting response and flash write data on the bus from when the Amiga CPU does a read.



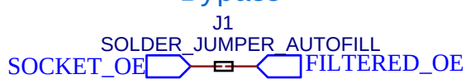


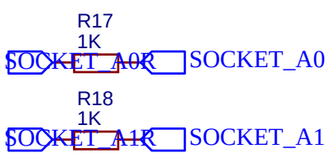
Required for Kicksmash1200
if not present, then must short
SOCKET_OE to FILTERED_OE.



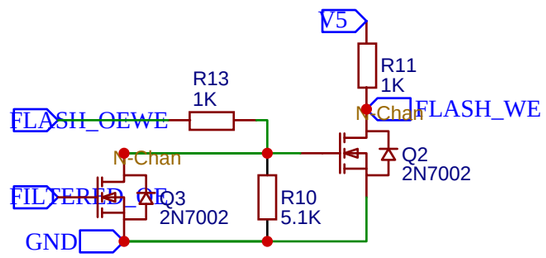


Bypass



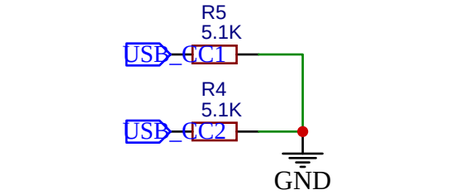


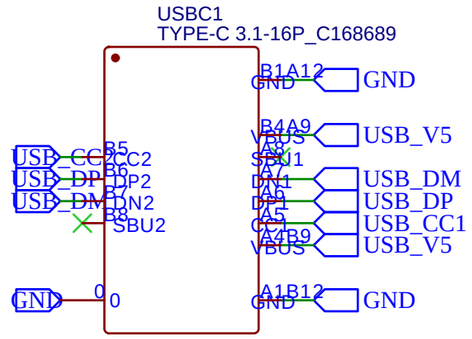
When FLASH_OEWE is high, FLASH_WE will be driven low when SOCKET_OE is low. Otherwise, FLASH_WE is managed by the STM32. This will allow the Amiga CPU to terminate a flash command write at the end of a bus cycle.



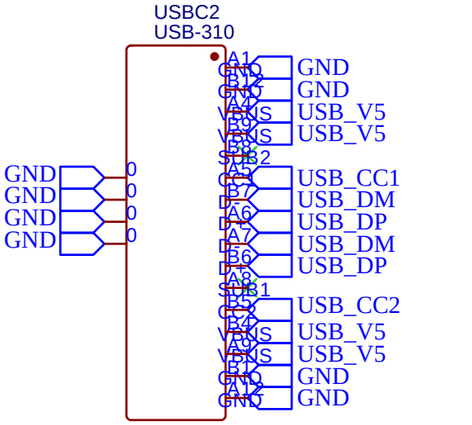
USB Type-C

Cutout version





Raised version

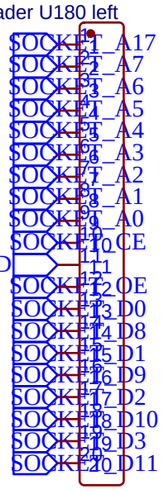


TITLE: Buffer + USB		REV: 10.1
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	Date: 2026-10-06 Drawn By: cdhooper	

A3000/A4000/4KT AA3K/4KCR

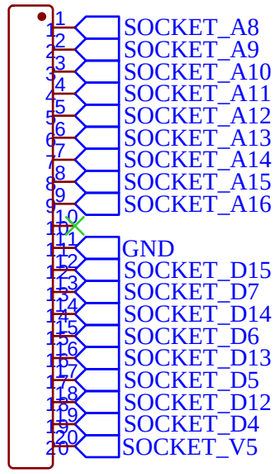
H1L

Bottom male header U180 left



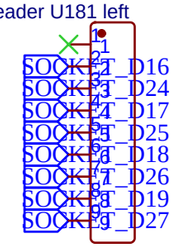
H1R

Bottom male header U180 right



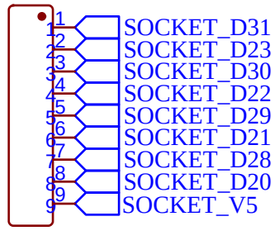
H2L

Bottom male header U181 left



H2R

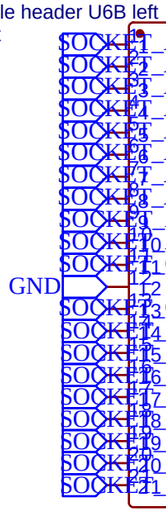
Bottom male header U181 right



A1200

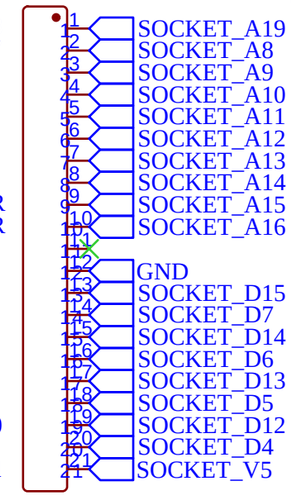
H5L

Bottom male header U6B left



H5R

Bottom male header U6B right

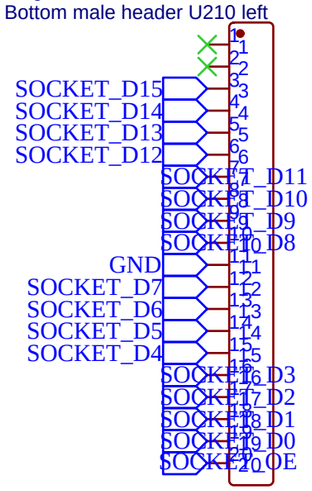


<-- A1200 also uses

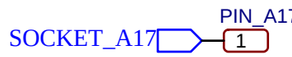
AmigaPCI AT27C4096 and A3000 U182/U183 ROM Tower

H3L

Bottom male header U210 left

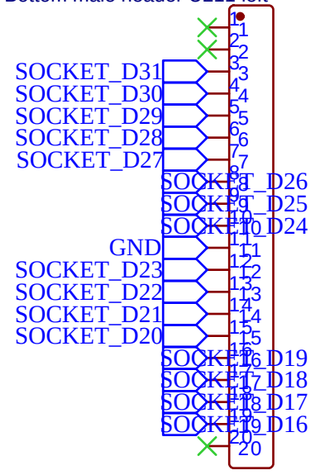


Right Socket D0-D15



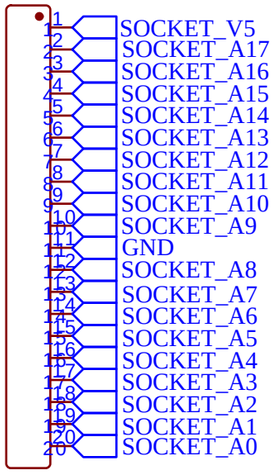
H4L

Bottom male header U211 left

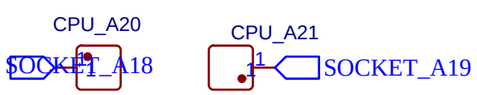


H4R

Bottom male header



Left Socket D16-D31



TITLE: Sockets		REV: 10.1
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	Date: 2026-10-06 Drawn By: cdhooper	